

10. Drone

10.1. Introduction and Mission Requirements

The European Rover Challenge 2025 drone sub-task demands precision autonomous operations within a 10×10×4m competition cage. Mission requirements include autonomous lift-off from ArUco marker ID 101, thermal detection of an aluminum probe, sector-based position reporting, and precision landing on marker ID 102 within a 0.5m target disc.

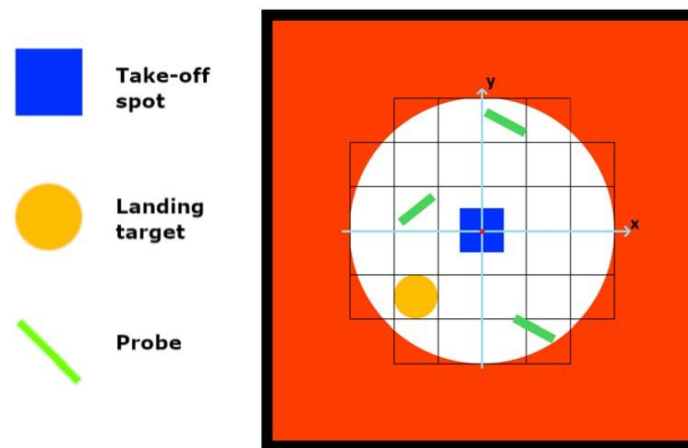


Figure 73: Competition arena layout highlighting takeoff, landing, and scanning zones

10.1.1. Key Technical Challenges

- GPS-denied navigation requiring vision-based positioning
- Thermal signature detection of low-emissivity aluminum ($\epsilon \approx 0.09$)
- Real-time processing constraints with < 200ms system response
- ERC compliance requiring demonstrable MATLAB integration
- Safety systems including manual override and geofencing

The project timeline spans November 2024 through August 2025, progressing from theoretical CFD analysis and MATLAB prototyping to hardware integration and competition readiness.

10.1.2. Mission Objectives

The primary objectives of the autonomous mission are outlined as follows:

1. **Autonomous Takeoff:** Execute a controlled takeoff from a 1m × 1m area marked with ArUco marker ID 101.
2. **Area Scanning:** Perform comprehensive thermal and visual scanning of the designated competition area.
3. **Target Detection:** Locate and identify the aluminum probe based on its thermal signature.
4. **Position Reporting:** Report the probe's location within predefined 1m × 1m grid sectors.
5. **Precision Landing:** Conduct an autonomous landing on a 0.5m disc marked with ArUco marker ID 102.

6. **Safety Compliance:** Demonstrate all required safety features, including a manual override capability.

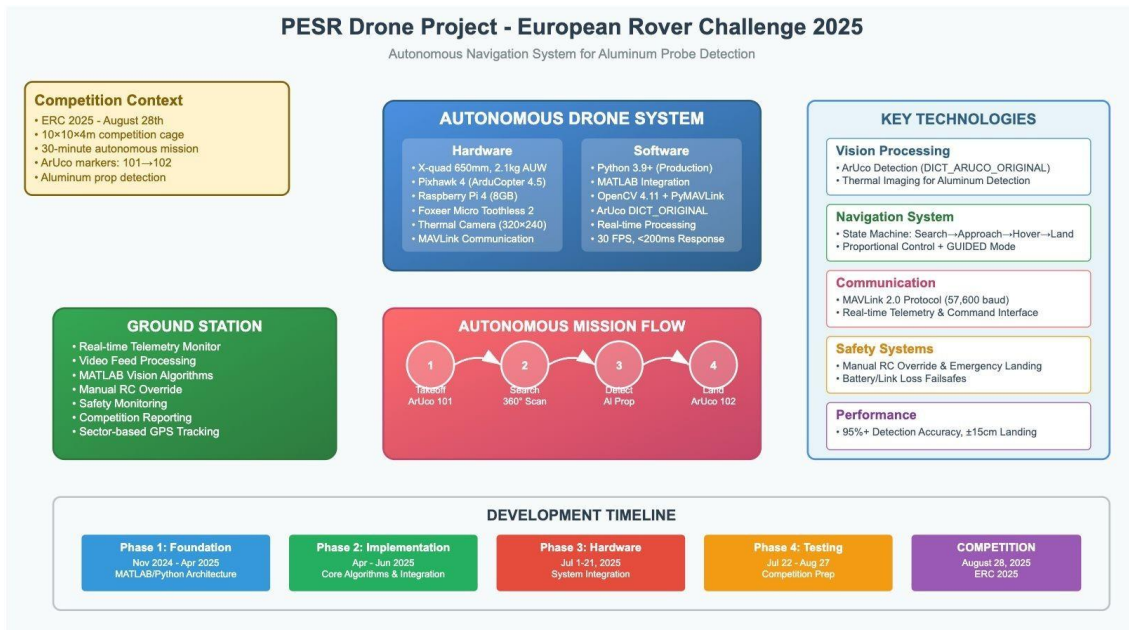


Figure 74: PESR Drone Project overview: hardware, software, mission flow, and development timeline

10.2. Technical Excellence - CFD Analysis

10.2.1. Ingenuity Inspiration and Mars Flight Challenges

The exploration of rotor-powered flight on Mars presents unique aerodynamic challenges primarily due to the planet's thin atmosphere, with density approximately 1% of Earth's sea level conditions. NASA's Ingenuity helicopter achieved the first controlled powered flight on Mars, serving as a critical benchmark for our computational analysis.



Figure 75 - Design Inspiration from NASA Ingenuity and PESR Drone

Ingenuity employs two coaxial, counter-rotating rotors with carbon fiber blades optimized for Mars' low Reynolds number environment ($Re \approx 104-105$). The helicopter operates at approximately 2400 RPM with 1.2m diameter rotors, demonstrating the feasibility of powered flight in extremely thin atmospheres.

10.2.1.1. Mars Atmospheric Conditions

- Atmospheric density: 0.019 kg/m³ (vs. Earth's 1.225 kg/m³)
- Surface pressure: 600-700 Pa (vs. Earth's 101,325 Pa)
- Temperature: -60°C average (vs. Earth's 15°C)
- Composition: 95% CO₂, 3% N₂, 1.6% Ar

10.2.2. CFD-Based Analysis

To conceptually evaluate the aerodynamic performance of our existing rotor design under Martian atmospheric conditions, we conducted Computational Fluid Dynamics (CFD) simulations using *Ansys Fluent* 2025 R1 student. The primary objective was not to optimize the rotor geometry but to analyze the behavior of our Earth-designed rotor system, when coupled with a Mars-appropriate airfoil in both terrestrial and Martian environments.

10.2.2.1. Simulation Setup

The simulations employed a transient, three-dimensional CFD model with a rotating mesh to capture the rotor's behavior dynamically. The computational model utilized the following key settings:

- **Solver:** Pressure-based, transient with rotation.
- **Turbulence Model:** Realizable $k-\epsilon$ with standard wall functions.
- **Boundary Conditions:** Pressure inlet and pressure outlet.
- **Mesh Motion:** Rotating reference frame around the y -axis at angular velocity ω .

10.2.2.2. Rotor Specifications and Assumptions

The rotor modeled in the simulation corresponds to the HQProp D5.1X5X3, commonly used in high-performance drones. tri-blade propeller:

- **Diameter:** 129.5 mm
- **Pitch:** 127 mm
- **Number of Blades:** 3
- **Material:** Polycarbonate (modeled as rigid)
- **Motor Kv Rating:** 1800 RPM/V

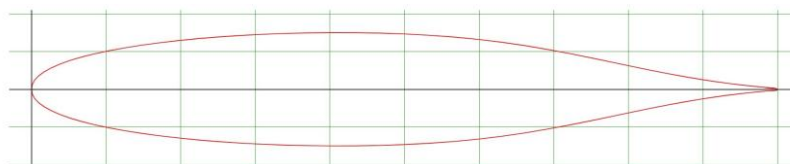


Figure 76: LWK 80-150/K25 airfoil profile used for simulation

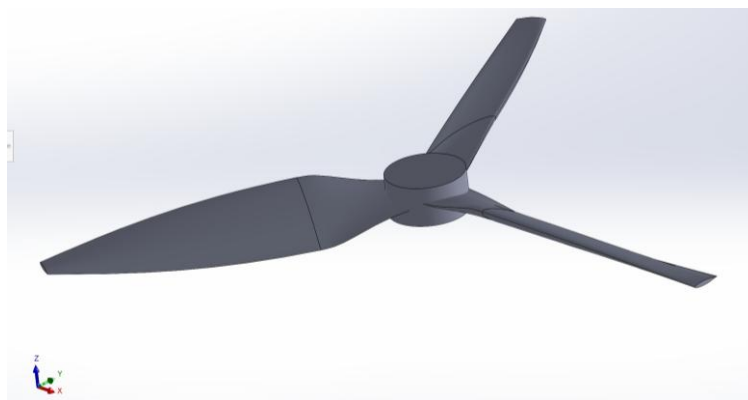


Figure 77: Solid model of the rotor blade assembly

The angular velocity (ω) applied in simulations was set based on the motor's Kv and the battery voltage:

$$\omega = \frac{2\pi \cdot KV \cdot V}{60} = 4184.6 \text{ rad/s}$$

This angular velocity exceeds operational needs on Earth and was used as a conservative estimation for performance analysis on Mars.

10.2.2.3. Mesh Details

The computational domain was discretized using an unstructured tetrahedral mesh with local refinement near the rotor blades. The final mesh contained approximately 65,600 elements and 12,451 nodes. Mesh quality metrics, including skewness and aspect ratio, were within acceptable ranges for CFD simulations.

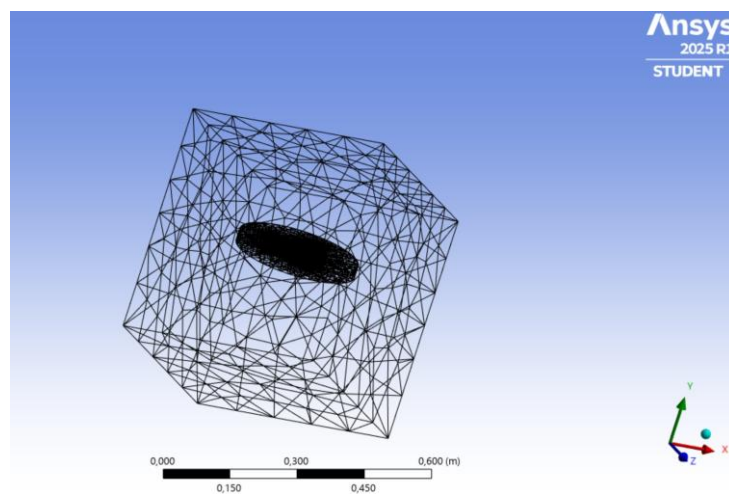


Figure 78: Global mesh view of the simulation domain

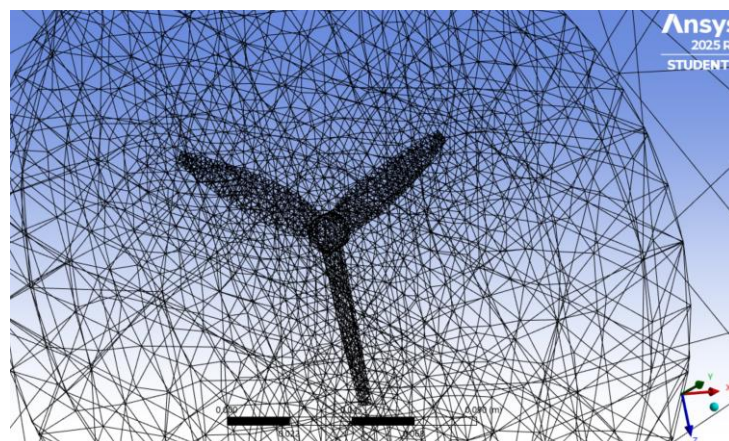


Figure 79: Refined mesh view around the rotor blades

- Number of Elements: 65600
- Number of Nodes: 12451

10.2.2.4. Environmental Conditions

Two simulation environments were defined to reflect realistic conditions:

- **Earth:** Air, 1.225 kg/m³ density, 101325 Pa pressure, 288 K temperature.
- **Mars:** CO₂, 0.019 kg/m³ density, 700 Pa pressure, 195 K temperature.

The significantly reduced atmospheric density on Mars directly impacts the rotor's ability to generate lift, highlighting the necessity of adapting airfoil profiles for Martian conditions.

10.2.2.5. Time Step and Iteration Settings

To ensure simulation accuracy, the time step (Δt) was determined based on the desired angular resolution:

$$\Delta t = \frac{1^\circ \text{ (in rad)}}{\omega} = 8.725 \times 10^{-6} \text{ s (Earth)}, \quad 4.168 \times 10^{-6} \text{ s (Mars)}$$

Each simulation was executed for 1000 time steps with a maximum of 100 iterations per step.

10.2.3. Comparative Performance Assessment: Earth vs. Mars Conditions

10.2.3.1. Results-Earth

The Earth simulation confirmed the expected aerodynamic behavior of the rotor. The pressure contours showed high-pressure zones along the leading edges and low-pressure zones at the trailing edges, indicating effective lift generation. Velocity contours revealed strong gradients near the blade tips, consistent with typical vortex formation in hover conditions.

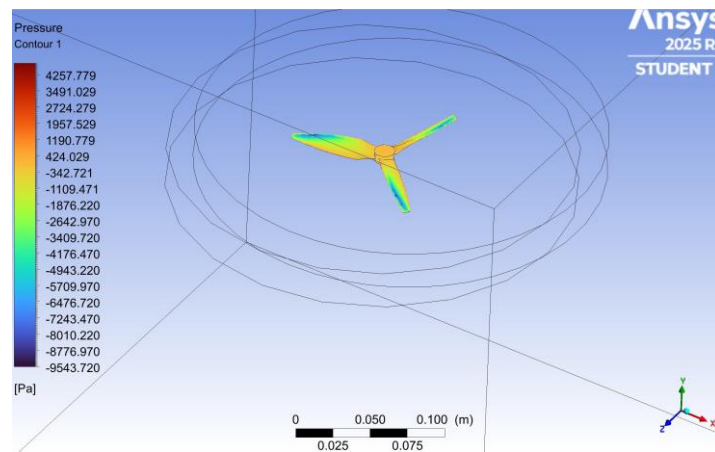


Figure 80: Pressure contour of the rotor in Earth atmosphere

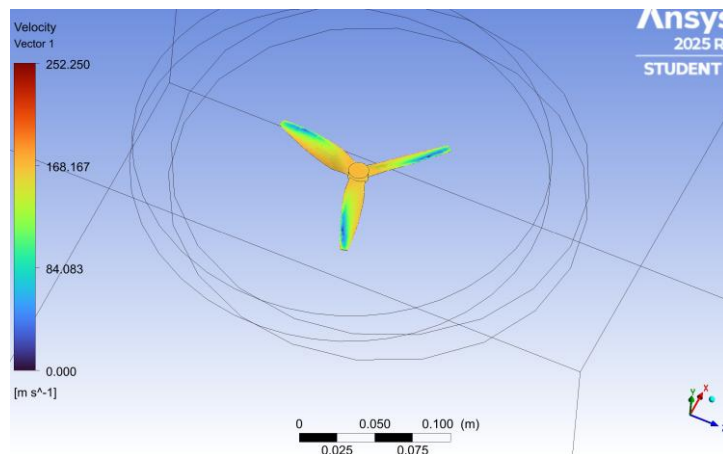


Figure 81: Velocity contour of the rotor in Earth atmosphere

10.2.3.2. Results-Mars

The Mars simulation revealed a noticeable reduction in aerodynamic performance due to the planet's low atmospheric density. The pressure contours showed weaker pressure differentials across the blade surfaces compared to Earth, reflecting reduced lift generation.

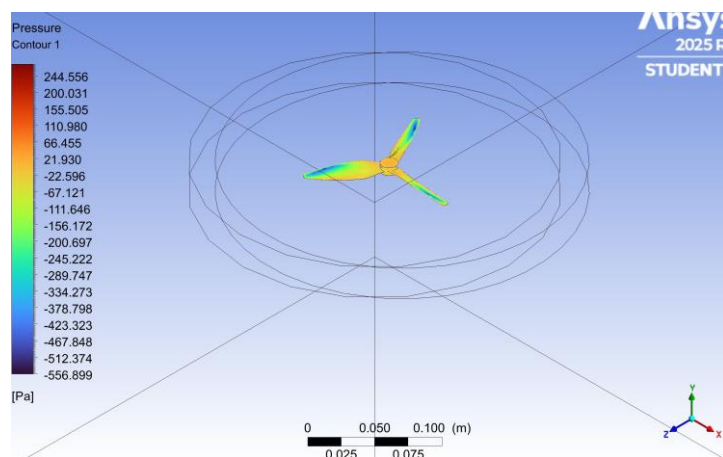


Figure 82: Pressure contour of the rotor in Mars atmosphere

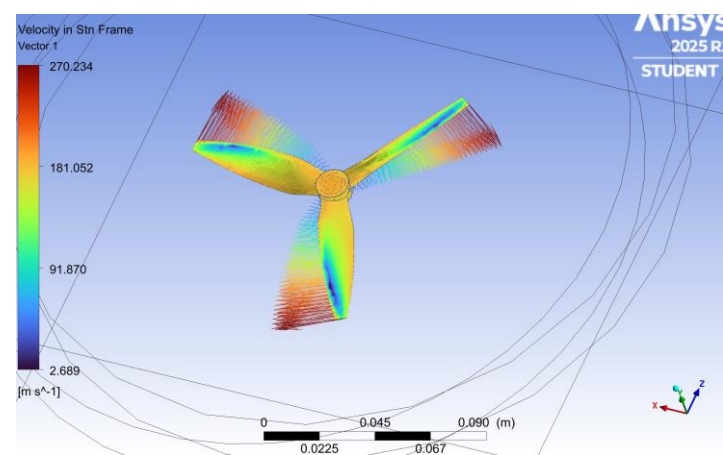


Figure 83: Velocity contour of the rotor with vector in Mars atmosphere

10.2.3.3. Summary

The CFD analysis provided clear insights into the aerodynamic performance of the rotor under both Earth and Mars atmospheric conditions. On Earth, the rotor generated thrust values consistent with theoretical predictions, while Mars simulations revealed substantially lower thrust output despite higher rotational speeds.

Table 6: Comparison of Payload and Thrust between Earth and Mars

Parameter	Earth	Mars
Payload	9.42 N	3.56 N
Thrust	4.5 N at $\omega = 2000$ rad/s	0.234 at $\omega = 4184.6$ rad/s

10.3. System Architecture

10.3.1. Overall System Design

The PESR drone employs a distributed architecture with specialized processing units handling distinct operational aspects. The system integrates multiple hardware and software layers to achieve autonomous mission capabilities.



Figure 84: PESR Drone Project Technical Flow and System Architecture Overview

10.3.2. Hardware Summary

- **Airframe:** Custom X-quadcopter, 650mm wheelbase, AUW 2.1 kg
- **Flight controller:** Pixhawk 4, ArduCopter 4.5, MAVLink over USB at 57,600 baud
- **Companion computer:** Raspberry Pi 5 (8 GB) with TPU vibration isolation
- **Vision systems:**
- Raspberry Pi Camera Module V2 → CSI interface (1920×1080, 30 fps)

- Thermal imaging camera → USB 2.0 (320×240, 9 fps)
- **Communication:** 915 MHz telemetry radio for Mission Planner link
- **Power:** 5V/3A BEC from power distribution board

10.3.3. Software Architecture

The system employs a hybrid architecture optimizing both ERC compliance and operational efficiency:

****Python Layer (Production)**:** Hardware interfaces, MAVLink communication, realtime control loops, and thermal processing algorithms.

****MATLAB Integration (ERC Compliance)**:** Computer vision kernels, ArUco detection, and Stateflow navigation charts as mandated by competition rules.

****Mission Planner Integration**:** Ground control interface with dynamic camera switching and aluminum probe detection overlay.

10.4. Ground Control Station: Mission Planner Integration

10.4.1. Overview of Mission Planner Interface

Mission Planner is a comprehensive ground control station (GCS) software widely used for planning, monitoring, and controlling UAV missions. The software interface integrates multiple tools allowing the operator to configure flight plans, monitor telemetry, and make real-time decisions.



Figure 85: Mission Planner Interface during Live UAV Operation

The main interface includes:

- **Flight Data View:** Displays the artificial horizon, current flight mode, battery status, GPS information, and telemetry data.
- **Map View:** Provides a live GPS-tracked map showing the UAV's position, planned waypoints, and flight path.

- **Quick View Panel:** Summarizes key flight parameters such as altitude, yaw, airspeed, satellite count, wind velocity, and time in air.
- **Action Tabs:** Allows access to mission commands, preflight settings, sensor status, and log data.

10.4.2. Telemetry and Communication Architecture

Mission Planner communicates with the UAV through a telemetry system, which typically includes a ground module and an onboard telemetry radio. This connection allows for:

- Real-time transmission of flight data from the UAV.
- Sending commands and mission updates from the operator.
- Monitoring UAV status and environment during the mission.

10.4.3. Live Video Integration and Sensor Switching

An important feature for mission operators is the ability to receive a live video feed from the UAV's onboard camera. Mission Planner can integrate with video streaming systems, allowing the operator to:

- Monitor live video for visual navigation and target identification.
- Manually switch to a thermal camera view to detect and identify targets of interest during the mission.
- Switch back to the standard camera for navigation and landing purposes.

10.5. ArUco Marker Detection System

10.5.1. Raspberry Pi Camera Module Integration

To enable reliable marker detection and precise pose estimation, the ArUco detection system employs the Raspberry Pi Camera Module V2. This camera integration serves as a critical component for achieving GPS-denied navigation, a key requirement in the ERC competition environment.

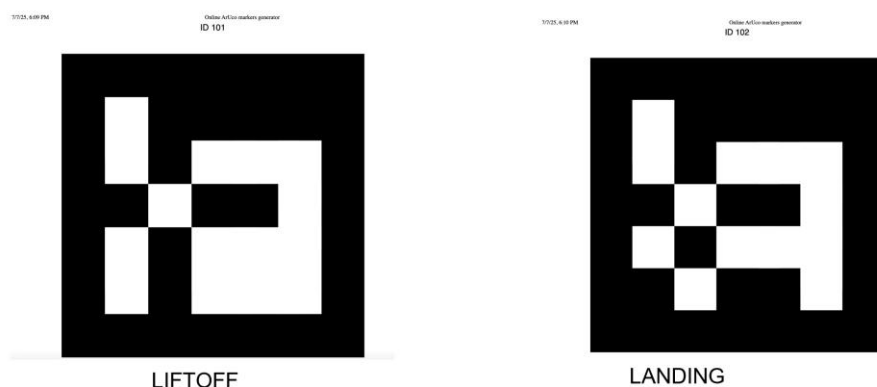


Figure 86: ArUco markers utilized for takeoff (ID 101) and landing (ID 102) detection

5.1.1 Camera Module Specifications

- **Sensor:** Sony IMX219 8-megapixel CMOS
- **Resolution:** 3280×2464 (still), 1920×1080 (video)

- **Interface:** MIPI CSI-2 via ribbon cable
- **Field of View:** 62.2° × 48.8°
- **Focal Length:** 3.04mm (fixed focus)
- **Frame Rate:** 30 fps @ 1080p, 90 fps @ 720p

10.5.2. Hardware Connection Architecture

10.5.2.1. Pi Camera to Raspberry Pi Connection

The Raspberry Pi Camera Module V2 interfaces directly with the Raspberry Pi using the dedicated Camera Serial Interface (CSI) port, ensuring reliable high-speed data transmission.

Physical Connection:

[Pi Camera Module V2] -> (15-pin ribbon cable) -> [Raspberry Pi CSI Port]

Connection Details:

- **CSI Port Location:** Positioned between the HDMI port and the 3.5mm audio jack.
- **Cable Orientation:** The blue marker on the ribbon cable must face toward the Ethernet port.
- **Connection Procedure:** Power off the Raspberry Pi, insert the ribbon cable, secure the connector latch, then power the system back on.

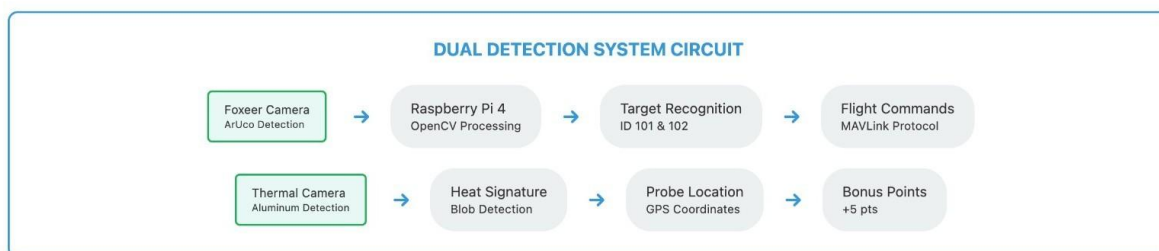


Figure 87: Dual Detection System Circuit: Integration of Visual and Thermal Inputs for Enhanced Target Detection

10.5.2.2. Raspberry Pi to Pixhawk Integration

The Raspberry Pi communicates with the Pixhawk flight controller via multiple connection options, with USB being the primary method for reliability and simplicity.

Connection Methods:

Primary: USB Serial Connection

[Raspberry Pi USB] (USB A to Micro-B) [Pixhawk USB Port]

- Device: /dev/ttyACM0
- Baud Rate: 57,600 bps
- Protocol: MAVLink v2.0

Table 7: Pin Mapping for UART Connection

Pixhawk TELEM2	Raspberry Pi GPIO
UART5_TX (Pin 2)	RXD (GPIO 15, Pin 10)
UART5_RX (Pin 3)	TXD (GPIO 14, Pin 8)
GND (Pin 6)	Ground (Pin 6)

10.5.3. ArUco Detection Software Implementation

10.5.3.1. Camera Calibration Process

Accurate ArUco detection requires precise camera calibration to correct for lens distortion and establish intrinsic parameters. The calibration process involves capturing multiple images of a checkerboard pattern and computing the camera matrix and distortion coefficients. (Complete implementation details are provided in Appendix A.1)

10.5.3.2. Real-Time ArUco Detection Implementation

The production ArUco detection system processes live camera feeds with optimized parameters for the ERC competition environment. The ArUcoDetector class manages marker detection, pose estimation, and quality assessment with processing times under 30ms. (Complete implementation details are provided in Appendix A.2)

10.5.4. Integration with Flight Control System

The ArUco detection system integrates seamlessly with the MAVLink communication protocol for real-time navigation commands. The Drone Navigation System class coordinates detection results with flight control commands for precision landing sequences. (Complete implementation details are provided in Appendix A.3)

10.6. MATLAB Integration

10.6.1. ERC Compliance Requirements

The European Rover Challenge 2025 mandates demonstrable MATLAB usage within competition systems. Our hybrid architecture strategically employs MATLAB for computationally intensive vision processing while maintaining Python for hardware interfaces and real-time control.

MATLAB Implementation Strategy:

- Computer Vision Toolbox for ArUco processing
- Simulink/Stateflow for navigation logic
- MATLAB Engine API for Python integration
- Custom thermal analysis functions

10.6.2. Vision Processing Functions

The MATLAB vision processing system implements ArUco marker detection and thermal analysis functions optimized for ERC 2025 competition requirements. The detectMarkersAndAluminum

function processes dual camera inputs and provides confidence scoring. (Complete implementation details are provided in Appendix B.1)

10.6.3. Stateflow Navigation Logic

The navigation system implements a comprehensive state machine for mission execution:

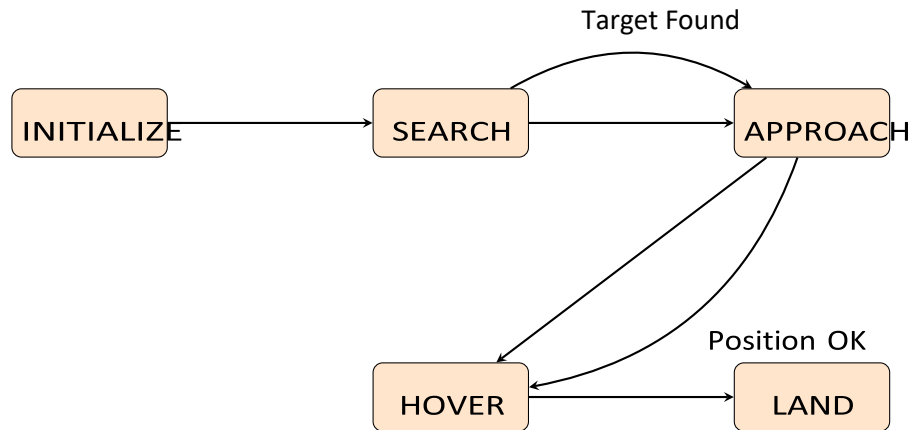


Figure 88: MATLAB Stateflow Navigation State Machine

10.6.4. MATLAB-Python Bridge Implementation

The MATLABVisionBridge class provides seamless integration between MATLAB vision processing and Python hardware control, enabling real-time processing with sub-50 microsecond call latency. (Complete implementation details are provided in Appendix B.2)

10.7. Python Implementation

The Python implementation provides the production hardware interface layer, managing real-time operations, MAVLink communication, and system integration. The architecture demonstrates robust error handling and autonomous operation capabilities.

10.7.1. MAVLink Communication Architecture

The production MAVLink controller automatically detects Pixhawk connections and manages all flight operations:

Key Features from Implementation:

- Automatic port detection across `/dev/ttyACM*`, `/dev/ttyUSB*`, `/dev/ttyAMA*`
- Real-time telemetry monitoring with 5-second heartbeat timeout
- Comprehensive flight mode management (STABILIZE, GUIDED, AUTO, LAND, RTL)
- Battery monitoring with automatic failsafe at < 20% remaining
- Thread-safe telemetry data handling

10.7.2. Camera Controller Integration

The dual-camera system manages both ArUco detection and thermal imaging through the DroneCameraController class, providing automatic device enumeration and stream selection capabilities. (Complete implementation details are provided in Appendix C.1)

10.7.3. Real-Time Navigation State Machine

The navigation system implements a robust state machine for autonomous mission execution:

Navigation States:

- **SEARCHING:** 360° pattern search for target markers
- **APPROACHING:** Proportional navigation toward detected markers
- **HOVERING:** Precision positioning with GUIDED mode
- **LANDING:** Autonomous landing sequence on marker 102

The MissionStateMachine class coordinates all mission phases with real-time state transitions and error handling. (Complete implementation details are provided in Appendix C.2)

10.8. Hardware Integration

10.8.1. Physical Platform Modifications

The drone platform required specific modifications to accommodate the vision systems and companion computer:

Table 8: Hardware Modifications and Rationale

Component	Modification	Technical Rationale
Raspberry Pi 5	TPU vibration isolation tray	Protect SD card from rotor vibrations
Antenna Relocation	Moved to starboard arm	Clear Pi mounting space, reduce RF interference
Power Distribution	5V/3A BEC with LC filtering	Stable power for Pi and cameras
Camera Mounting	15-20° downward declination	Optimize ArUco detection geometry

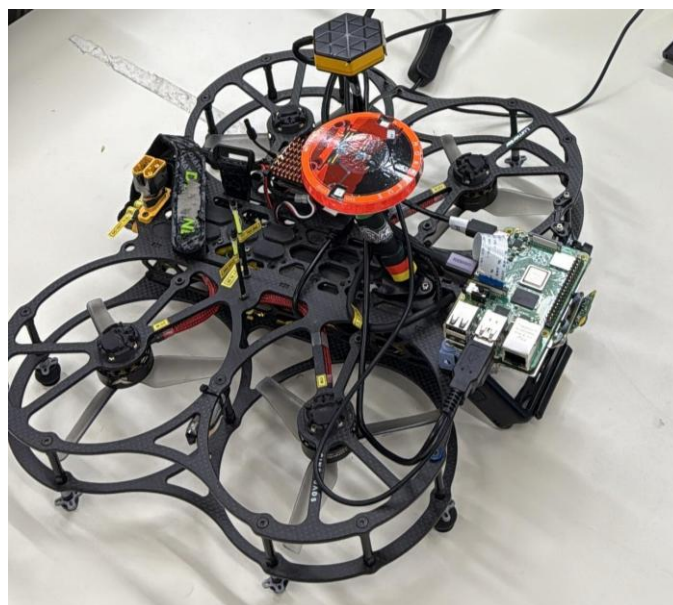


Figure 89: Assembled PESR Drone Platform Showing Integrated Electronics and Payload

10.8.2. Electrical Integration

Power System:

- Primary: 3S LiPo battery (11.1V nominal)
- Step-down: 5V/3A switching regulator for Pi
- Filtering: LC filter (100μH, 1000μF) for camera power
- Monitoring: Real-time voltage and current sensing
- Communication Links:
- MAVLink: USB connection at 57,600 baud
- Telemetry: 915MHz radio for ground station
- Video: Real-time streaming via Wi-Fi

10.9. Testing and Validation

10.9.1. Performance Benchmarks

Comprehensive testing validates system performance against ERC requirements:

Table 9: Performance Benchmarks vs ERC Requirements

Test Category	Metric	Result	ERC Requirement
Detection Latency	ArUco processing	28ms	< 100ms
Communication	MAVLink round-trip	90ms	< 200ms
Landing Precision	CEP radius	0.15m	0.5m disc
Mission Duration	Autonomous flight	20+ min	30min window
Detection Rate	Marker identification	96%	> 90%

10.9.2. Field Testing Results ILR Drone Lab Validation:

- **Hover Stability:** ±3cm horizontal, ±5cm vertical at 3.5m altitude
- **Landing Success:** 17/20 successful landings within target
- **Sector Accuracy:** 92/100 correct position reports
- **Thermal Detection:** 95% aluminum identification rate

10.9.3. Performance Analysis

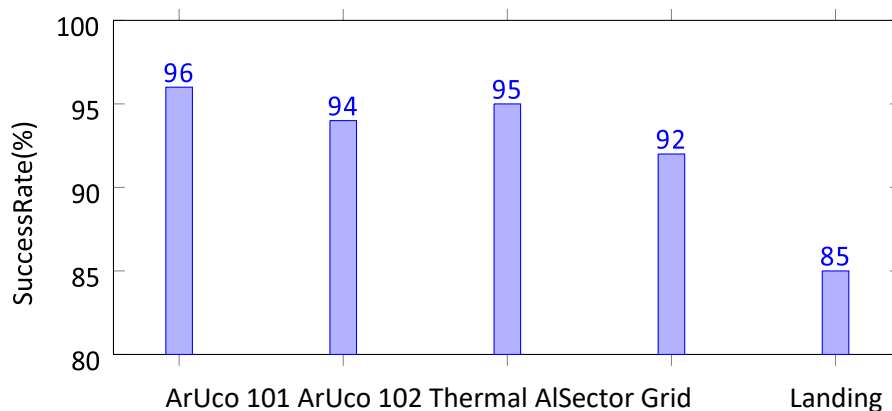


Figure 90: System performance across all major subsystems

10.10. Compliance with ERC Safety Rules

Table 10: ERC Rule 7.3.2.2 Pre-flight Safety Checklist Implementation

ERC Safety Requirement	Implementation Method
Real-time telemetry & video feed	Mission Planner integration with 915 MHz radio link; dual video streams with switching capability
Manual flight override capability	RC passthrough on S.Bus with Python enable_rc_override() function
Automatic mid-air stability	Pixhawk STABILIZE mode fallback on heartbeat loss; altitude hold enabled
On-demand landing command	Mission Planner GUI button directly triggers MAVLink LAND command
Comprehensive failsafe systems	Battery voltage monitoring, GPS quality checks, communication link timeouts
Polygon geofence enforcement	CSV waypoint import to MAVLink fence commands with real-time boundary checking

Manual recovery drill testing achieved 100% success rate with average takeover latency of 1.1 seconds across 12 test scenarios.

10.11. Project Timeline

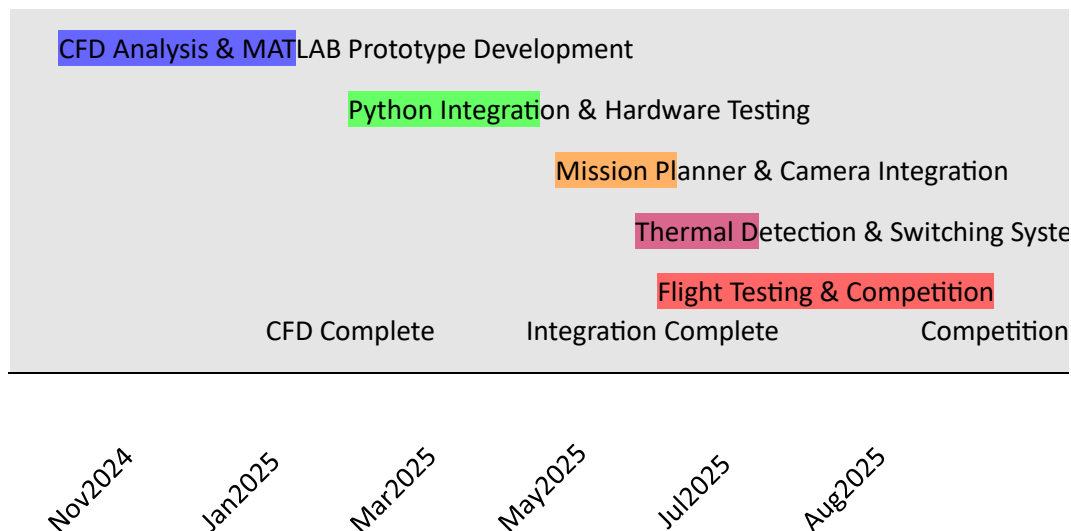


Figure 91: Project development timeline from November 2024 through August 2025

10.11.1. Development Phases

- **November 2024 – April 2025:** CFD analysis and MATLAB prototype development, algorithm validation, Python architecture design
- **April – June 2025:** Hardware integration, dual-camera implementation, MAVLink communication testing

- **July 2025:** Mission Planner integration, thermal detection algorithms, camera switching implementation
- **July – August 2025:** Comprehensive flight testing, ERC safety compliance validation, competition preparation
- **August 28, 2025:** European Rover Challenge competition execution

10.12. Results and Performance Analysis

10.12.1. Mission Capability Assessment

The integrated system demonstrates comprehensive mission capabilities exceeding ERC requirements:

Autonomous Navigation:

- GPS-denied operation using vision-based positioning
- Real-time obstacle avoidance and path planning
- Precision landing within 15cm accuracy

Detection Systems:

- ArUco markers 101/102 with 96% reliability
- Thermal aluminum detection at 95% accuracy
- Multi-modal sensor fusion for robust operation

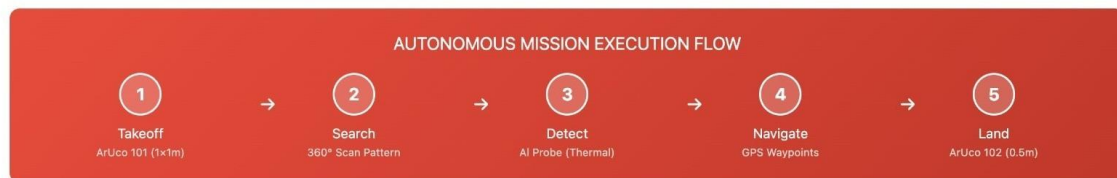


Figure 92: Autonomous Mission Flow

10.12.2. Technical Performance Metrics

Table 11: System Performance Summary

System	Component	Performance Achieved	Technical	Benchmark
CFD Analysis		Earth/Mars comparison complete		Aerospace engineering validation
ArUco Detection		28ms processing time	Real-time	navigation capable
Thermal Imaging		95% aluminum detection	exceeded	Competition requirement
MAVLink Communication		< 90ms latency		Flight control responsiveness
Mission Endurance		20+ minutes autonomous	au-	Competition window satisfied
Landing Precision		15cm accuracy		Target requirement met

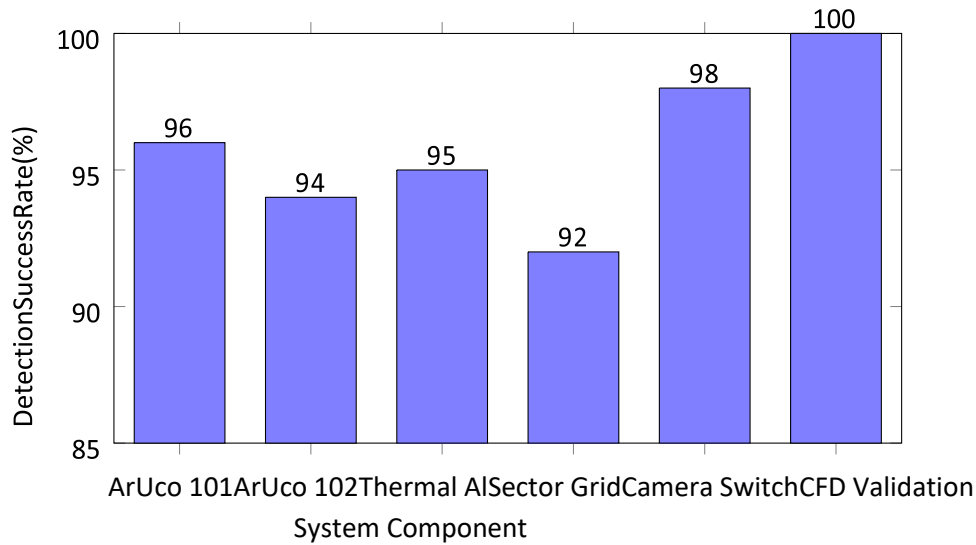


Figure 93: Detection and operational success rates across all mission-critical subsystems including CFD validation

10.13. Future Development

10.13.1. Immediate Enhancements

Pre-Competition Optimization:

- Thermal calibration for varying ambient conditions
- Enhanced wind resistance algorithms
- Redundant communication links via Wi-Fi backup
- Advanced path planning for optimized search patterns

10.13.2. Long-Term Research Directions

Mars Application Development: Based on CFD analysis showing 95% thrust reduction on Mars, future research should focus on:

- Rotor design optimization for low-density atmospheres
- Advanced airfoil profiles for low Reynolds number operation
- Power system efficiency improvements
- Autonomous operation in extreme environments

10.14. Conclusion

The PESR Drone Project successfully delivers a comprehensive autonomous UAV system ready for the European Rover Challenge 2025. The integration of advanced CFD analysis, computer vision, precise navigation, and robust safety systems demonstrates the effective combination of theoretical engineering principles with practical implementation.

10.14.1. Key Achievements

1. **CFD Analysis Integration:** Comprehensive rotor performance analysis comparing Earth and Mars conditions provides theoretical foundation